CPEN 355 - Final Project

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December 14, 2023

1 Introduction

This project aims to investigate the defining features of three different species of penguins using the Palmer Penguins Extended Dataset found on Kaggle. The objective is to train a classification model on this dataset to predict whether a penguin is an Adelie, Gentoo, or Chinstrap penguin. The dataset provides samples of penguins with corresponding attributes such as bill length, flipper length, sex, and diet. For simplification, the scope of this project is limited to healthy, adult penguins.

2 Exploratory Data Analysis

The Palmer Penguins Extended Dataset contains eleven features columns. Two of the feature columns, life stage and health metrics, will not be used since the scope of this project will be limited to adult, healthy penguins. The remaining nine are described below:

- **Species:** Species of the penguin (Adeli, Gentoo, Chinstrap)
- **Island:** Island where the penguin was found (Biscoe, Dream, Torgensen)
- **Sex:** Gender of the penguin (male, female)
- **Diet:** Primary diet of the penguing (fish, krill, squid)
- **Year:** Year the data was collected (2021-2025)
- **Body Length (mm):** Bill length in millimeters
- **Bill Depth (mm):** Bill depth in millimeters
- **Flipper Length (mm):** Flipper length in millimeters
- **Body Mass (g):** Body mass in grams

Filtering the dataset to only include healthy, adult penguins, the penguins dataset contains 362 samples: 159 samples are Adelie, 130 samples are Gentoo, and 73 samples are Chinstrap. None of these samples have missing values.

Inspecting the feature columns, it seems that the dataset contains a mix of numerical columns (bill length, bill depth, flipper length, body mass), categorical columns (island, diet, year), and binary features (sex).

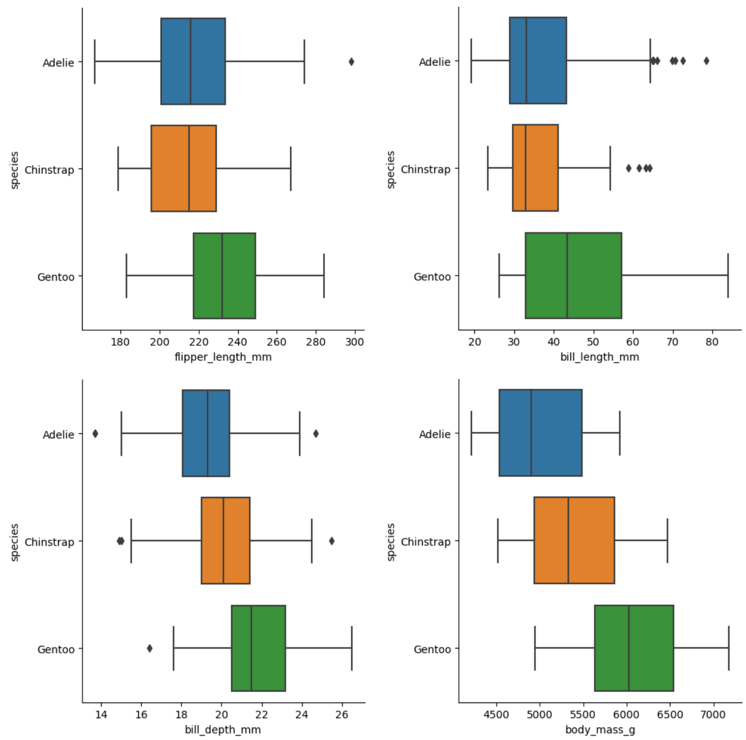


Figure 1: Box Plots of Numeric Features by Penguin Species

The numerical features were first explored to determine general patterns and relationships that can be found between the three species of penguins. In Figure 1, the difference in flipper length, bill length, bill depth, and body mass between the Adelie, Gentoo, and Chinstrap was analyzed using box plots.

- **Flipper Length:** Gentoo penguins, on average, seem to have the longest bills. Adelie penguins and Chinstrap penguins seem to have shorter bill lengths, but between these two species their bill lengths are quite similar. Note that the data spread range for the three species are very similar and that the dataset contains class imbalance, so it is difficult to establish a definite relationship.
- **Bill Length:** Gentoo penguins, on average, seems to have the longest bills. Adelie penguins and Chinstrap penguins seem to have similar bill lengths, with Adelie penguins having a larger spread of bill lengths than Chinstrap.
- **Bill Depth:** Gentoo penguins, on average, seem to have the largest bill depth. Adelie penguins have the smallest bill depth on average.
- **Body Mass:** Gentoo penguins, on average, seem to have the largest body mass. Adelie penguins have the smallest body mass on average.

Next, the categorical features and their relationship to the penguin species were explored. Figure 2 shows the percentage of a penguin species found on each island. It was found that all Chinstrap penguins were found on Dream Island and all Gentoo penguins were found on Biscoe Island. However, Adelie penguins were found on all three islands and the percentage of Adelie penguins found on each island were quite evenly distributed.

Figure 2: Percentage of Penguin Found on Each Island by Species

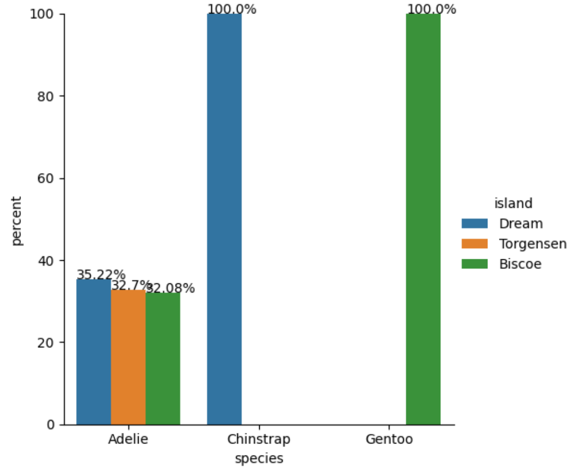
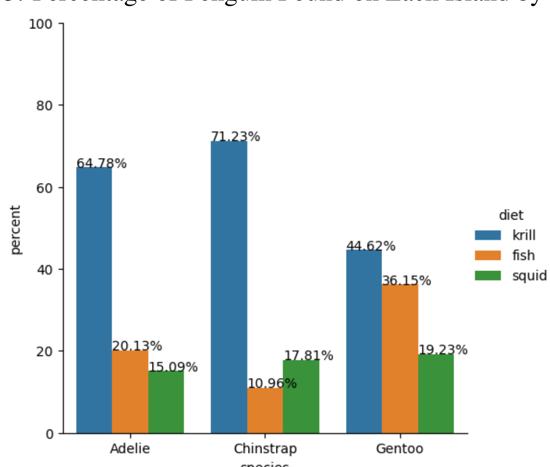


Figure 3 shows the percentage of penguins for each type of primary diet by species. Krill seems to be the most popular diet for all three penguin species, the Chinstrap species having the largest percentage of penguins having krill as their primary diet (71.23%) with Adelie following closely in second (64.78%). Compared to the other species, Gentoo has the highest percentage of penguins that consume fish as their primary diet (36.15%) whereas Chinstrap has the lowest percentage (10.96%). The percentage of penguins that consume squid as their primary diet is the most similar across all species, with Gentoo also having the highest percentage (19.23%) compared to other species, and Adelie having the lowest (15.09%)

Figure 3: Percentage of Penguin Found on Each Island by Species



Finally, the trends between numeric features (i.e. flipper length, bill length, bill depth, body mass) and the year was briefly explored to see if there were any significant changes to the penguins within the five years. Table 1 shows the mean of each numeric feature by year.

While there are slight fluctuations in penguin features over the years, the changes are minimal and do not seem to follow a trend. They do not seem to be substantial. This consistency could imply stable environmental conditions or successful adaptation strategies over the observed years.

Table 1: Cross-Validation Scores for Baseline Model

Year	Flipper Length (mm)	Bill Length (mm)	Bill Depth (mm)	Body Mass (g)
2021	228.2	44.5	20.6	5520.6
2022	223.5	38.5	20.7	5538.5
2023	220.2	39.7	20.6	5475.0
2024	223.1	41.9	20.1	5452.1
2025	222.7	38.6	20.1	5402.5

3 Model Selection

The chosen machine learning model to predict the species of a penguin was Random Forest Classifier. Since task of predicting between three different species is a multi-class classification, Random Forest is well-suited for such problems. It is also capable of capturing non-linear relationships, which ecological datasets like the penguins datset often involves. Finally, Random Forests often produce high training scores and are not prone to overfitting since they average the predictions of multiple decision trees, making them robust to noise in the data.

4 Data Splitting & Preprocessing

Since the penguins dataset is a relatively small dataset, it was split following a 80/20 train-to-test ratio. This is to maximize the size of the training set while ensuring the test set is large enough to provide a reliable evaluation of the machine learning model.

Moving on to the preprocessing stage, the year feature column was first removed as the changes in the penguin attributes over the years are not substantial and descriptive of the penguin species. The training data was then transformed accordingly based on their feature type, which is summarized in Table 2. The goal is to prepare the data appropriately for a Random Forest Classifier model.

Table 2: Features Types for Penguin Dataset

Numeric Feature	bill length, bill depth, flipper length, body mass
Categorical Feature	island, diet
Binary Feature	sex

Numeric features were standardized using sklearn's **StandardScaler**, which removes the mean and scales the numeric features to unit variance. Categorical features were transformed using sklearn's **OneHotEncoder** in order to convert them into a numerical format for the machine learning model to process and to avoid introducing ordinality. Binary features were also transformed using **OneHotEncoder** with the binary option in order to avoid multicollinearity. Finally, using sklearn's **ColumnTransformer** the appropriate transformations were applied to each feature in a structured method.

5 Model Training

Using sklearn's **RandomForestClassifier**, a Random Forest Classifier model with a very small number of trees with small depths (n_estimators = 10 and max_depth = 5) was initially trained to predict the penguin species in the test dataset. This is to establish a baseline performance of the model before moving on to hyperparameter tuning. A pipeline was created using sklearn's

Pipeline to combine the preprocessing steps and the model training process into a single object. This simplifies the fitting and predicting process and ensures that the data is treated consistently.

Table 3: Cross-Validation Scores for Baseline Model

Fold	Cross-Validation Score
1	0.776
2	0.845
3	0.897
4	0.862
5	0.877

Table 3 shows the 5-fold cross-validation scores using the baseline model. The scores are good and expected for this Random Forest Classifier baseline, however they are not consistent at each fold and vary significantly within the range 0.776 to 0.897. This may be concerning and could indicate that the validation sets have varying distributions, leading to discrepancies in the cross-validation scores. The train dataset may be too small or the class imbalance between the different penguin species may be too large, or both.

Predicting the penguin species in the test dataset, the test accuracy score was found to be 0.822, as shown in Table 4. This score is fairly close to the mean cross-validation score, which is 0.851.

Table 4: Accuracy Scores for Baseline Model

Mean Cross-Validation Score	0.851
Test Score	0.822

Given the large variation in cross-validation score at each fold, the test accuracy score may not be representative of the model's performance. Therefore, the precision and recall for each penguin species were also analyzed.

Table 5: Classification Report for Baseline Model

	Precision	Recall
Adelie	0.83	0.68
Gentoo	0.69	0.73
Chinstrap	0.88	1.00

Table 5 shows that the Chinstrap class has the highest precision at 0.88. This means that the model correctly predicted a penguin in the test dataset as a Chinstrap penguin 88% of the time. Surprisingly, the recall for the Chinstrap class is perfect at 1.00. This means that all Chinstrap penguins in the test dataset have been identified correctly. This shows that the baseline model is already good at identifying a Chinstrap penguin.

The precision for the Adelie class is also relatively good, however its recall is quite low at 0.68. In contrast, the Gentoo class has both low precision and recall. For the next part, hyperparameter tuning will be performed to attempt to improve the overall model performance, especially for predicting the Adelie and Gentoo classes.

Hyperparameter tuning was performed using sklearn's **GridSearchCV** on the Random Forest Classifier hyperparameters n_estimators and max_depth. The search space was defined to be n_estimators = (50, 100, 120, 140, 160, 180, 200) and max_depth = (5, 10, 15, 20, 25). As a result of grid search, the best n_estimators was found to be 180 and the best max_depth was found to be 10.

6 Model Prediction and Evaluation

With the best optimal values for n_estimators and max_depth, a new Random Forest Classifier model was trained using these optimal parameters.

Table 6: Cross-Validation Scores for Random Forst Classifier

Fold	Cross-Validation Score
1	0.862
2	0.845
3	0.879
4	0.931
5	0.912

Table 6 shows the 5-fold cross-validation scores. The optimal parameters increased the overall cross-validation scores of the Random Forest Classifier. They also improved the variation in the cross-validation scores, but not at a significant level.

Table 7: Accuracy Scores for Random Forest Classifier

Mean Cross-Validation Score	0.886
Test Score	0.904

Using this new model to predict the penguin species in the test dataset, the test accuracy score increased to 0.904, as shown in Table 7. This score is a little higher than the mean cross-validation score, which along with the varying cross-validation scores indicates that the model's performance cannot be reliably captured by the test accuracy. Therefore, the precision and recall for this optimal Random Forest Classifier were analyzed in Table 8.

Table 8: Classification Report for Random Forest Classifier

	Precision	Recall
Adelie	0.86	0.89
Gentoo	0.92	0.73
Chinstrap	0.94	1.00

The precision for the Adelie class improved only slightly, however the recall improved significantly from 0.68 to 0.89. With the optimized model, 89% of the Adelie penguins in the test set were identified correctly. The precision for the Gentoo class also significantly improved from 0.69 to 0.92. The model correctly predicted a penguin as an Gentoo penguin 92% of the time, as opposed to the initial 69%. However, the recall for the Gentoo class did not improve at all and remained at 0.73. Similarly, the recall for the Chinstrap class remained at 1.00 and the precision for the Chinstrap class improved from 0.88 to 0.94.

7 Conclusion

This project aimed to develop a machine learning model to accurately predict the species of penguins—Adelie, Gentoo, or Chinstrap—using the Palmer Penguins Extended Dataset. A Random Forest Classifier was chosen for its suitability for multi-class classification tasks, its ability to capture complex, non-linear relationships in the data, and its robustness against overfitting.

The initial Random Forest Classifier model, considered as the baseline model, demonstrated promising performance. However, variability in cross-validation scores indicated potential issues with the model's consistency. Further hyperparameter tuning using GridSearchCV led to the identification of optimal parameters, resulting in improved accuracy, precision, and recall.

The precision and recall metrics indicated the ability of the model to correctly identify each species, particularly for the Chinstrap penguins. The increase in these metrics for the Adelie and Gentoo species, after optimization, further validates the efficacy of the chosen model and the tuning process.

However, there are limitations to this model. The first limitation arises from the class imbalance within the dataset. Despite efforts to mitigate this through model choice and hyperparameter tuning, this imbalance could still have influenced the model's performance across the different species. Future work could explore more advanced resampling techniques or custom loss functions to further address class imbalance.

In summary, this project shows the potential of Random Forest Classifiers in ecological studies and species classification. While effective, the approach may benefit from a more extensive dataset, addressing class imbalances more thoroughly, and possibly exploring a wider array of advanced modeling techniques.